

Numpy_Basics

February 11, 2025

```
[6]: import numpy as np
import pandas as pd
```

```
[23]: a = np.array([[1,2,3],[4,5,6]])
print(a)
print(a.reshape(3,2))
```

```
[[1 2 3]
 [4 5 6]]
[[1 2]
 [3 4]
 [5 6]]
```

```
[20]: a = np.arange(6)
print(a)
```

```
[0 1 2 3 4 5]
```

```
[27]: a.ravel
a.shape
```

```
[27]: (2, 3)
```

```
[31]: a = np.arange(10)
```

```
[40]: print(a)
```

```
[0 1 2 3 4 5 6 7 8 9]
```

```
[41]: a.reshape(2,5)
```

```
[41]: array([[0, 1, 2, 3, 4],
          [5, 6, 7, 8, 9]])
```

```
[42]: a = a.reshape(5,2)
```

```
[43]: a
```

```
[43]: array([[0, 1],
          [2, 3],
```

```
[4, 5],  
[6, 7],  
[8, 9]])
```

```
[45]: a[:3]
```

```
[45]: array([[0, 1],  
           [2, 3],  
           [4, 5]])
```

```
[46]: a[-1]
```

```
[46]: array([8, 9])
```

```
[53]: a[-2,1:2]
```

```
[53]: array([7])
```

```
[54]: for row in a:  
       print(row)
```

```
[0 1]  
[2 3]  
[4 5]  
[6 7]  
[8 9]
```

```
[55]: for cell in a.flat:  
       print(cell)
```

```
0  
1  
2  
3  
4  
5  
6  
7  
8  
9
```

```
[56]: a
```

```
[56]: array([[0, 1],  
           [2, 3],  
           [4, 5],  
           [6, 7],  
           [8, 9]])
```

```
[59]: b = np.arange(5,12)
```

```
[66]: b = np.append(b,12)
```

```
[73]: b = b.reshape(4,2)
```

```
[74]: np.vstack((a,b))
```

```
[74]: array([[ 0,  1],  
          [ 2,  3],  
          [ 4,  5],  
          [ 6,  7],  
          [ 8,  9],  
          [ 5,  6],  
          [ 7,  8],  
          [ 9, 10],  
          [11, 12]])
```

```
[78]: a = np.delete(a,4,axis = 0)  
print(a)
```

```
[[0 1]  
 [2 3]  
 [4 5]  
 [6 7]]
```

```
[81]: mix = np.hstack((a,b))
```

```
[82]: mix
```

```
[82]: array([[ 0,  1,  5,  6],  
          [ 2,  3,  7,  8],  
          [ 4,  5,  9, 10],  
          [ 6,  7, 11, 12]])
```

```
[85]: even = mix % 2 == 0
```

```
[86]: even
```

```
[86]: array([[ True, False, False,  True],  
          [ True, False, False,  True],  
          [ True, False, False,  True],  
          [ True, False, False,  True]])
```

```
[87]: even = mix[even]
```

```
[89]: even.reshape(4,2)
```

```
[89]: array([[ 0,  6],
            [ 2,  8],
            [ 4, 10],
            [ 6, 12]])
```

```
[97]: print(a)
```

```
[[0 1]
 [2 3]
 [4 5]
 [6 7]]
```

```
[96]: for x in np.nditer(a, order = 'C'):
        print(x)
```

```
0
1
2
3
4
5
6
7
```

```
[94]: for x in np.nditer(a, order = 'F'):
        print(x)
```

```
0
2
4
6
1
3
5
7
```

```
[102]: print(a)
        print()
        print(b)
```

```
[[0 1]
 [2 3]
 [4 5]
 [6 7]]
```

```
[[ 5  6]
 [ 7  8]
 [ 9 10]
 [11 12]]
```

```
[99]: for x,y in np.nditer([a,b]):  
      print(x,y)
```

```
0 5  
1 6  
2 7  
3 8  
4 9  
5 10  
6 11  
7 12
```

```
[ ]:
```